

LISP CG Tools for AutoCAD

By AKH

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<https://github.com/debear81/lispCGtools>

General Installation

Files available on GitHub Repository: <https://github.com/debear81/lispCGtools>

Copy all *.lsp and *.dcl files into a common folder. This tool kit requires the (4) following files to function together:

- CG_2D.lsp
- CG_3D.lsp
- CG_helpers.lsp
- CG_dialogs.dcl

Add folder to AutoCAD Support Paths.

If needed, load the following commands using APPLOAD to make them available for use:

- CG2D.lsp
- CG3D.lsp

CG2D Typical Workflow

CG2D is a command that will extract an object (polyline or region) cross-sectional information for one or more selected objects about the origin of the drawing's User Coordinate System (UCS) and calculate the properties about the object's local centroid.

1. Draw closed 2D polylines or regions
2. Select desired 2D objects
3. Run CG2D
4. Save CSV
5. Review exported results
6. Verify CG marker placement

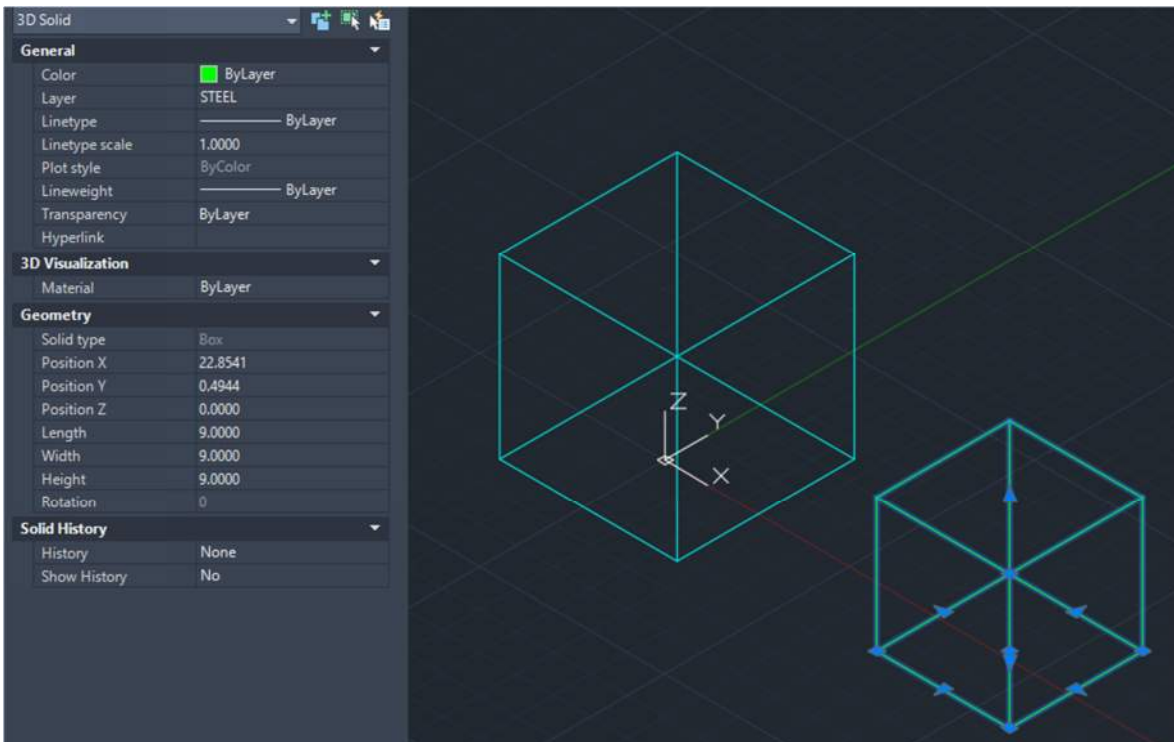
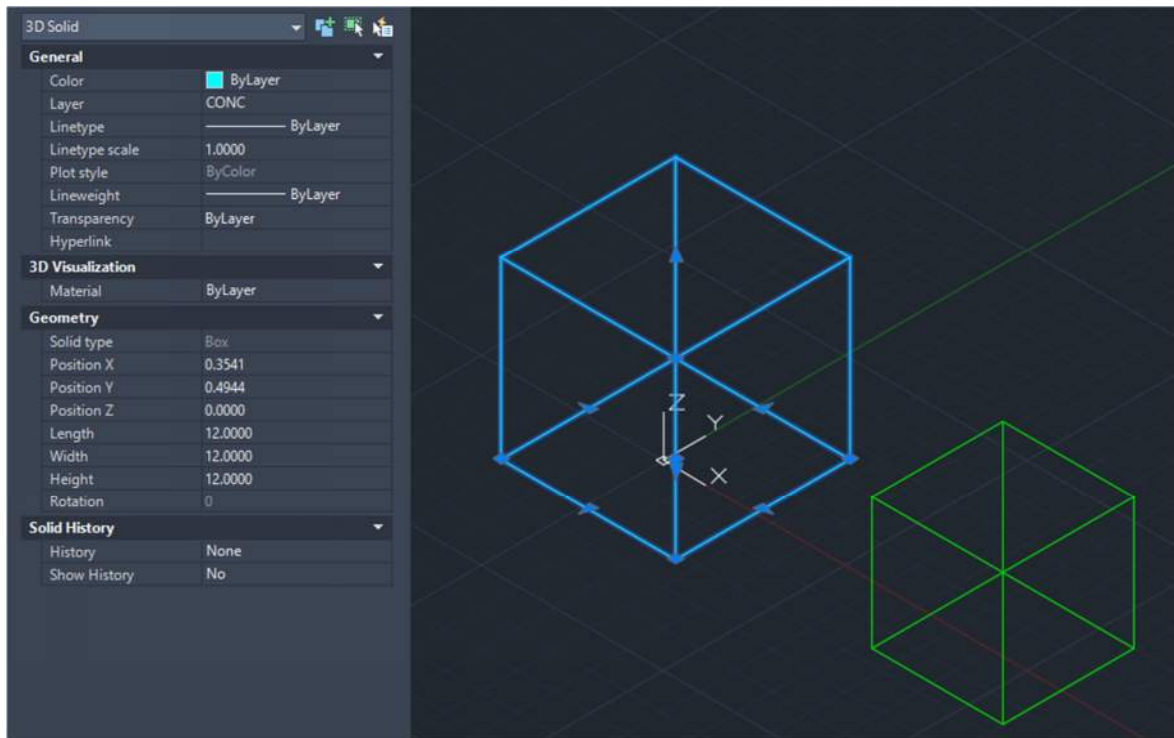
CG3D Typical Workflow

CG3D is a command that will calculate the combined centroid of one or more 3d solids in AutoCAD that represent one or more material types. The original idea for this stemmed from trying to automate & simplify preparing crane picks of non-uniform lifts for erection procedures.

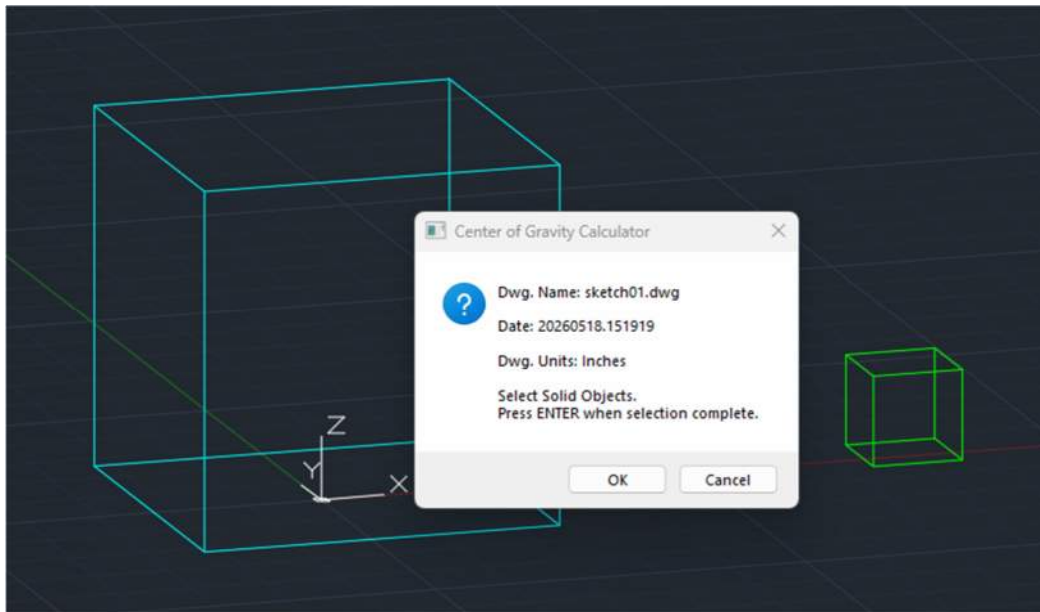
1. Model solids
2. Assign solids to layers by material
3. Run CG3D
4. Select solids
5. Enter densities
6. Save CSV
7. Review exported results
8. Verify CG marker placement

CG3D Command User Guide

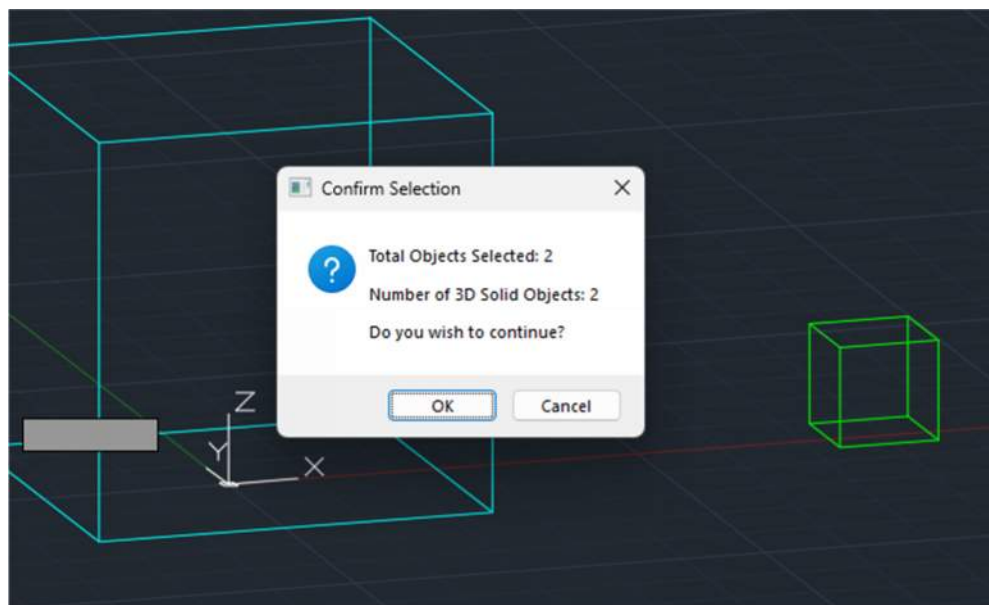
Layer names are used only for material-density assignment. Though it generally makes sense to give the specific layer a name that is based on the material name.



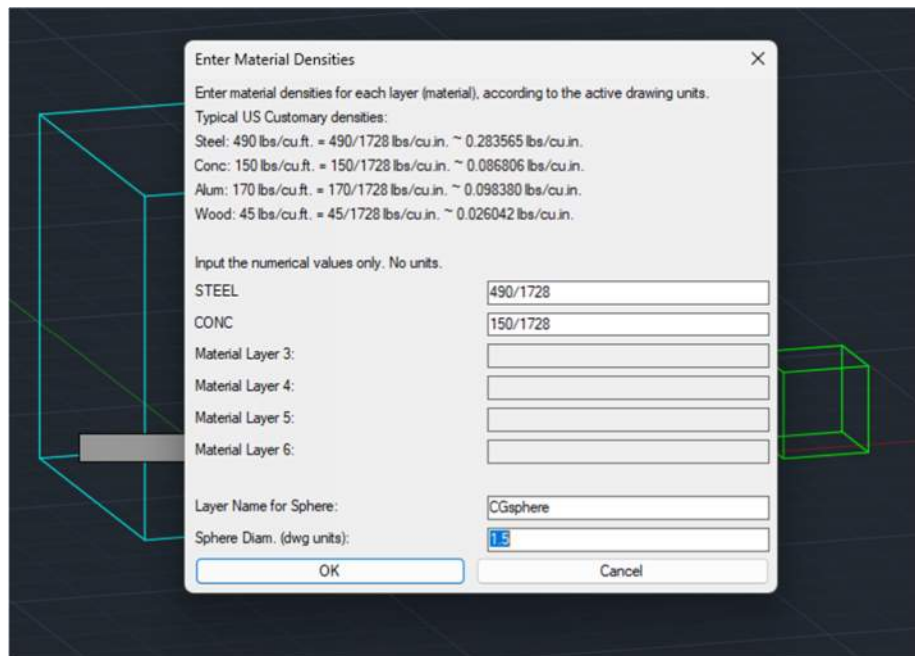
Enter the command “CG3D” and the drawing name, date, and working units are displayed for reference.



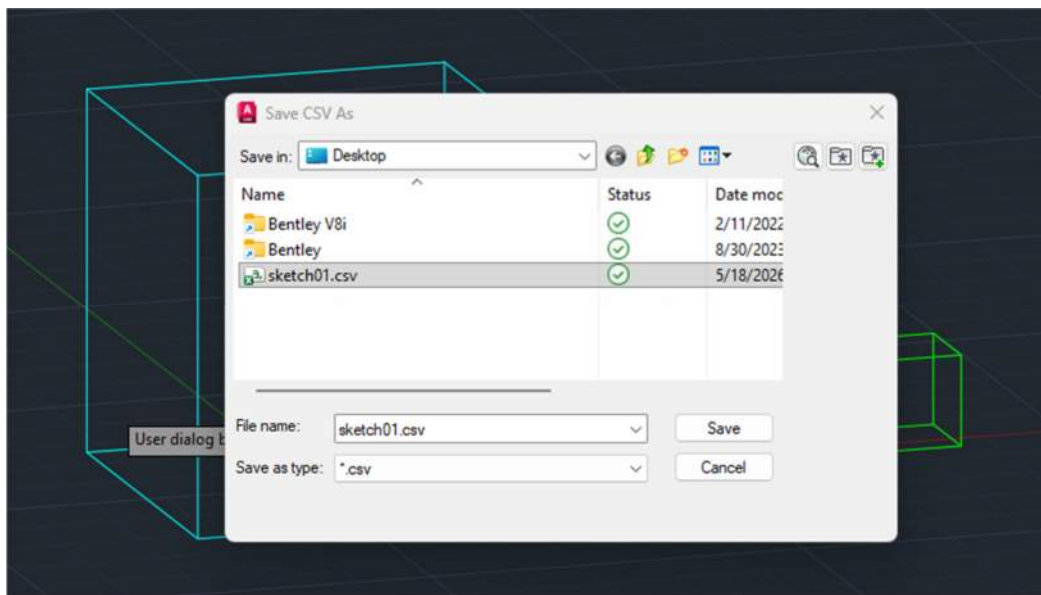
Once the desired objects (3Dsolds) are selected, the selection summary is displayed for verification.



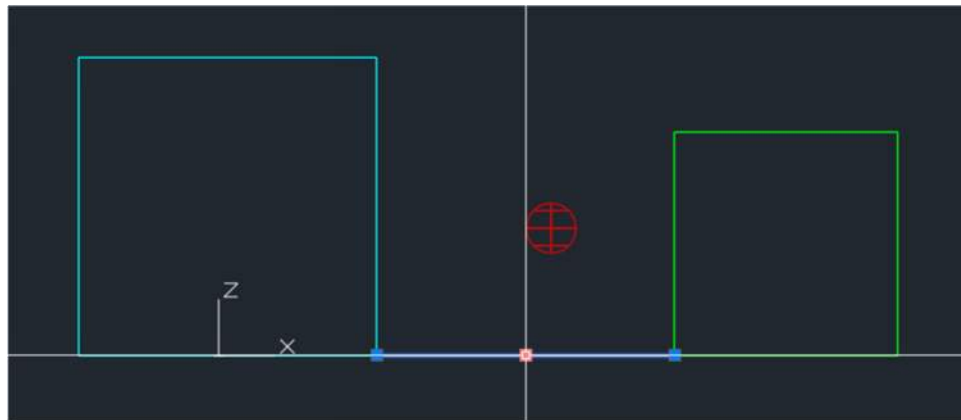
Next, a dialog box is displayed for entering material densities () that will be applied to the solids on each different layer. The user also has the opportunity to assign a specific name to the layer (that will be created, if necessary) on which a sphere representing the center of gravities location will be placed.



The user is requested to specify the name and location of the CSV file that will be created.



And the sphere representing the combined centroid is placed in the drawing for reference.



Then the CSV file can be imported into Excel as a nice table of AutoCAD information and calculated values.

	A	B	C	D	E	F	G	H	I	J	K	L
1	Dwg. Name:	sketch01.dwg										
2	Date:	20260518.15										
3	Units:	Inches										
4	ObjectID	ObjectName	Layer	Density	Volume	Centroid X	Centroid Y	Centroid Z	Weight	W * X	W * Y	W * Z
5	44	AcDb3dSolid	STEEL	0.283565	729	22.8541	0.494385	4.5	206.719	4724.37	102.199	930.234
6	43	AcDb3dSolid	CONC	0.086806	1728	0.354092	0.494385	6	150	53.1138	74.1577	900
7	SUM TOTALS											
8					Tot. Vol.				Tot. Weight	Tot. WX	Tot. WY	Tot. WZ
9					2457				356.719	4777.48	176.356	1830.23
10	CG COORDINATES											
11										Centroid X	Centroid Y	Centroid Z
12										13.3929	0.494385	5.13075

CG3D Current Capabilities

- Calculates combined CG of multiple 3DSOLIDs
- Supports multiple material densities
- Exports CSV summary
- Places CG marker sphere in model space
- Works with AutoCAD 2024

CG3D Current Limitations

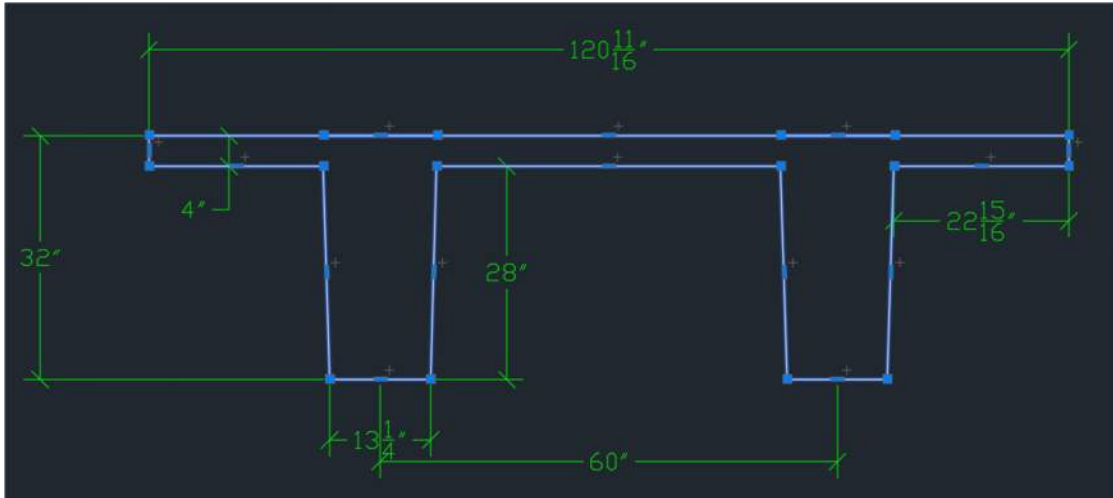
- 3DSOLIDs only
- Density units must match drawing units
- Does not support:
 - regions
 - surfaces
 - nested blocks
 - void subtraction
 - non-uniform material gradients

CG2D Command User Guide

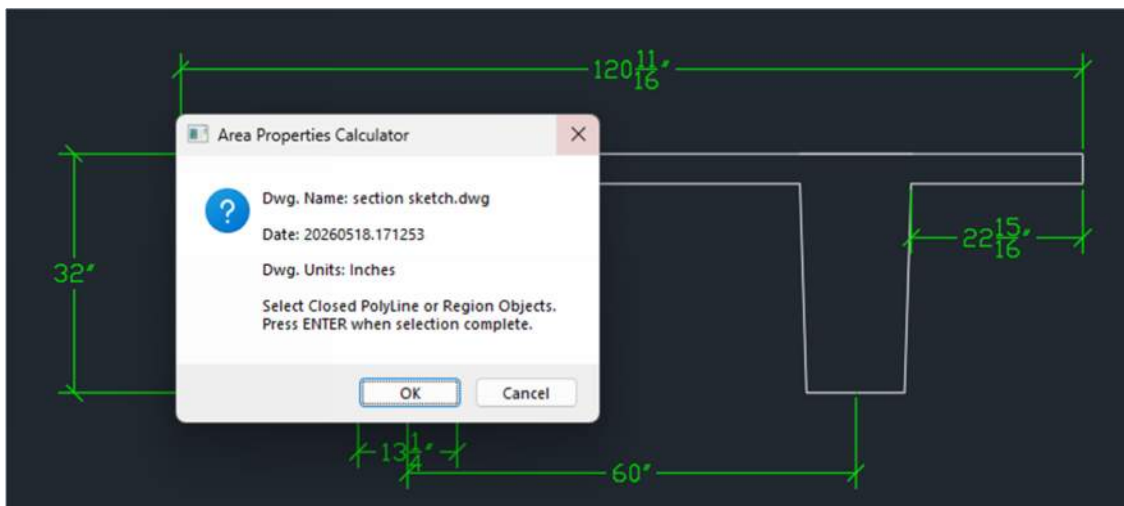
CG2D can be used with one or more closed shapes (polylines) for determining the local cross-sectional properties, but only calculates the properties for each individual shape. It does not currently calculate transformed or combined properties for multiple independent shapes.

CG2D is similar to the native MASSPROP command in that it provides cross-sectional properties; however, it automatically reports and exports properties relative to each shape's local centroid.

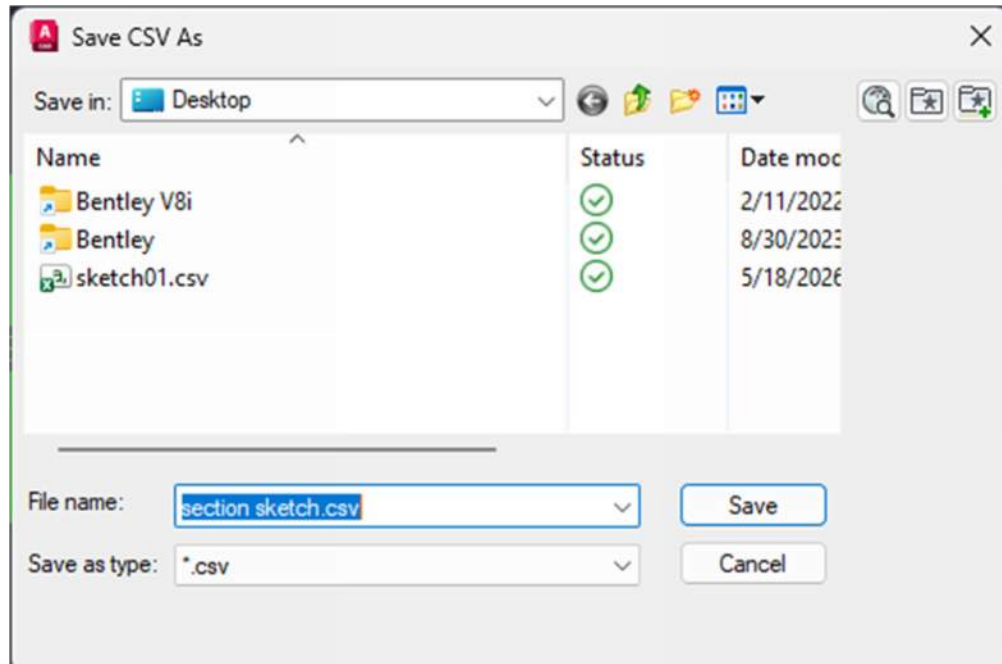
CG2D works with closed polylines and/or regions. So shapes drawn with individual lines and polylines need to be closed and joined using the PEDIT or MPEDIT command.



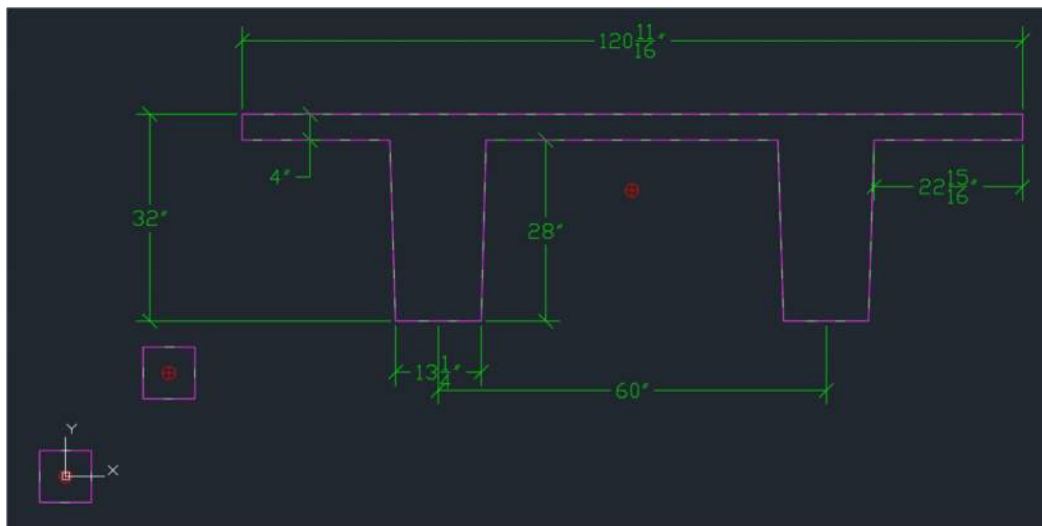
When the CG2D command is run, the pertinent drawing information is displayed and the user is directed to select desired objects.



Once OK is clicked, the routine proceeds and stores extracted and calculated data in a CSV file, which can be named and located by the user.



After the CSV is saved, circles/benchmarks are drawn at each shapes centroid and a region (purple in this image) is placed at each shape location. Note that multiple sections can be processed at the same time.



The CSV file can be opened and examined in Excel for verification.

	A	B	C	D	E	F	G	H	I	Q	R	S	T	U	V
1	Dwg. Name:	section sketch.dwg													
2	Date:	20260518.17													
3	Units:	Inches													
4															
5	ID	Name	Layer	Area	Perimeter	Cent X (global)	Cent Y (global)	Inert X (global)	Inert Y (global)	Inert X (lx, cg)	Inert Y (ly, cg)	Prod Inert (lxy, cg)	Rad Gyr X (rx, cg)	Rad Gyr Y (ry, cg)	Polar Inert (I, cg)
6	56	AcDbRegion	CG_TEMP_REGION	64	32	16	16	16725.3	16725.3	341.333	341.333	-32768	2.3094	2.3094	682.667
7	57	AcDbRegion	CG_TEMP_REGION	64	32	0	0	341.333	341.333	341.333	341.333	0	2.3094	2.3094	682.667
8	58	AcDbRegion	CG_TEMP_REGION	1268.43	358.016	87.5692	44.2811	2.61E+06	1.10E+07	125801	1.31E+06	-9.84E+06	9.95884	32.0765	1.43E+06

CG2D Current Capabilities

- Calculates individual CG of multiple closed polylines or regions
- Supports multiple closed sections
- Exports CSV summary
- Places CG marker in model space
- Works with AutoCAD 2024

CG2D Current Limitations

- Temporary working regions created by the routine are not automatically removed.
- Circle and ellipse entities have not been fully tested or implemented.
- Combined (unconnected) multi-shape sections have not been incorporated
- Void subtraction (holes) have not been incorporated
- Nested blocks have not been incorporated
- Composite sections and varying materials have not been incorporated

General Disclaimer

These routines are provided as-is for engineering assistance and drafting automation. The user is responsible for independently verifying all calculated values and suitability for use.